Revision-Note Review Worksheet for Expert Rater

This booklet contains one student revision note to review. It is a take-home revision note prepared by a teacher for a JSS1 Mathematics class on Three dimensional figures: properties of cubes, cuboids, pyramids and cones. Judge how well it covers the intended objectives and curriculum, follows the teacher's lesson plan, stays on topic and is conceptually correct, and is clear and useful for the students.

Lesson and revision-note context

Subject	Mathematics
Grade	JSS1
Term	Third Term
Week	4
Topic	Three Dimensional Shapes
Subtopic	Three dimensional figures: properties of cubes, cuboids, pyramids and cones
Theme	—
Curriculum objectives	• Identify the properties of cube and cuboids • Identify the properties of pyramids and cones
Previous knowledge	• Students have been taught the differences between cube and cuboid.
Lesson sequence	• Identify the properties of cube and cuboids — the teacher writes the properties of cubes and cuboids on the board; the student take notes in their notebooks • Identify the properties of pyramids and cones — the teacher draws the pyramids and the cone on the board and writes their properties; the student take them down, as in Step II
Instructional materials	• Shapes of cubes, cuboid, Pyramid, cone, cylinder and spheres.
Evaluation	• How many vertices, has a cuboid? • How many edges has a cube? • Draw a cube and cuboid then label. • Draw an hexagonal base pyramid and list the properties. • How many vertices has a tetrahedron.

Lesson objectives

1.	Identify the properties of cube and cuboids
2.	Identify the properties of pyramids and cones

Read the student revision note on the following pages, then complete the rating sheet at the end of this booklet.

STUDENT REVISION NOTE

Topic: Three dimensional figures: properties of cubes, cuboids, pyramids and cones

Class: JSS1 **Subject:** Mathematics **Week:** 4

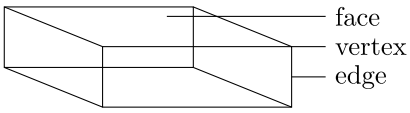
Section 1: Properties of Cubes and Cuboids

Three dimensional shapes are shapes that have length, breadth and height. Because they take up space in all three directions, they are also called solid shapes.

Solid shapes may be regular or irregular. Regular shapes are those with a regular, familiar form — a cube of sugar, a drink can, a match box. Irregular shapes are shapes without that regular form, like a stone or a tree.

Before we list the properties of a cube and a cuboid, here are three words you need. An edge is a line where two faces meet; it may be straight or curved. A face (or surface) may be flat or curved. A vertex (plural: vertices) is a point corner where edges meet. Look at fig-02: the leader line to the flat part is the face, the point corner is the vertex, and the line where two faces meet is the edge.

Visual



A cuboid drawn the same way, wide and shallow, with three leader lines to the right naming the top face, the nearer top-right corner and the nearer right-hand vertical edge.

A cube is drawn in fig-01. Look at it and count its parts:

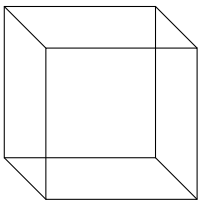
1. It is a solid shape.
2. It has 6 square faces.
3. It has 12 edges.
4. It has 8 vertices.

A cuboid is drawn in fig-02. Compare it with the cube:

1. It has 6 faces.
2. It has 12 edges.
3. It has 8 vertices.

The cube and the cuboid are the same in how many faces, edges and vertices they have. What makes them different is the shape of the faces: every face of a cube is a square, while the faces of a cuboid can be rectangles.

Visual



A cube drawn as a see-through solid: two equal squares, the nearer one offset down and to the right of the farther one, with the four pairs of corresponding corners joined.

Practice

1. How many vertices has a cuboid?
2. How many edges has a cube?
3. How many faces does a cube have, and what shape is each face?
4. In your own words, what is a vertex? Give an example from fig-02.

Section 2: Pyramids and Cones: Naming Them and Counting Their Parts

A pyramid is named after the shape of its base. A triangular base gives a triangular based pyramid, a square base gives a square based pyramid, a pentagonal base gives a pentagonal based pyramid, and a rectangular base gives a rectangular based pyramid. The apex is the single point above the base, and every other corner of the base is joined up to that apex.

Every pyramid is built from the same three parts:

Faces — the flat surfaces. A pyramid has its base plus one triangular face for each side of the base.
Edges — the lines where two faces meet.
Vertices — the corners where edges meet.

The table below shows the counts for the four pyramids you can see in the figures.

Shape	Faces	Edges	Vertices
Triangular based pyramid (tetrahedron)	4	6	4
Square based pyramid	5	8	5
Pentagonal based pyramid	6	10	6
Rectangular based pyramid	5	8	5

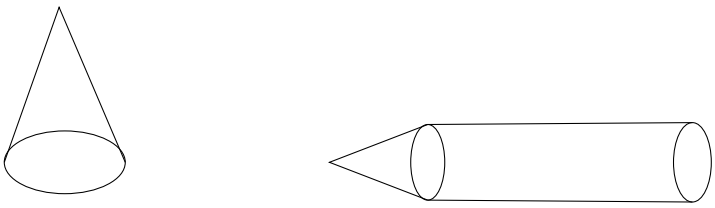
Notice the pattern: each time the base gains one more side, the faces, edges and vertices each go up by one.

The cone is a different shape from a pyramid. Its parts are:

Two faces — one curved face (the sloping side) and one flat face (the base).
One edge — the single circular line where the curved face meets the flat base.
One vertex — the apex, the point at the top.

So a cone has 2 faces, 1 edge and 1 vertex. You can see the upright cone and the cone lying on its side in fig-07.

Visual



Two drawings side by side. On the left an upright cone: an elliptical base with the two slant sides running up from the ends of the ellipse to the apex. On the right a cone lying on its side, apex at the left, its two slant sides reaching a narrow upright ellipse whose top and bottom then continue as two parallel lines to a second narrow ellipse further right.

Let's name and count the parts of a square based pyramid.

Name it. The base is a square, so the shape is called a square based pyramid.

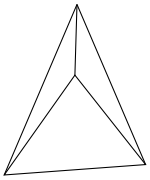
Count the faces. There is the square base, and one triangular face for each of the four sides of that base. So it has 1 base + 4 triangular faces = 5 faces.

Count the edges. The base has 4 edges, and each of the 4 triangular faces reaches up to the apex, adding 4 slanting edges. So it has 4 + 4 = 8 edges.

Count the vertices. There are 4 corners on the base plus the apex = 5 vertices.

So a square based pyramid has 5 faces, 8 edges and 5 vertices — exactly what the table says.

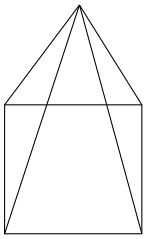
Visual



Triangular base Pyramid
(Tetrahedron)

A triangular base pyramid. The base triangle is drawn in perspective, its third vertex raised into the middle of the figure, and the apex above is joined to all three base vertices.

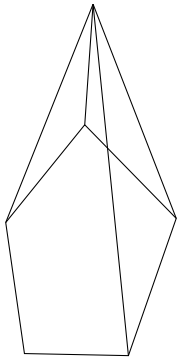
Visual



Square base Pyramid

A square base pyramid. The base is drawn as an upright rectangle and the apex above is joined to all four of its corners, so two edges run outside the rectangle and two cross it.

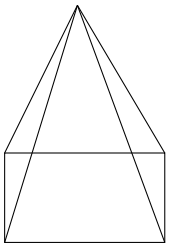
Visual



Pentagonal based
Pyramid

A pentagonal based pyramid. The base is drawn as a pentagon standing on one side with its fifth vertex uppermost, and the apex above is joined to that vertex, to the two vertices either side of it, and to the lower-right vertex — four edges from the apex, the last of them crossing the face of the solid.

Visual



Rectangular based
Pyramid

A rectangular based pyramid, drawn like the square one but on a base wider than it is tall.

Practice

1. How many vertices does a tetrahedron (a triangular based pyramid) have?
2. Name a pyramid whose base has six sides, and list its faces, edges and vertices.
3. List the faces, edges and vertices of a cone.

Check your work

Attempt every question before you read this page.

Properties of Cubes and Cuboids

1. A cuboid has 8 vertices.
2. A cube has 12 edges.
3. A cube has 6 faces, and each face is a square.
4. A vertex is a point corner where edges meet. In fig-02 the point corner where three edges meet is a vertex.

Pyramids and Cones: Naming Them and Counting Their Parts

1. 4 vertices

   A tetrahedron has 3 corners on its triangular base plus the apex, so 3 + 1 = 4 vertices.
2. It is called an hexagonal based pyramid. It has 7 faces, 12 edges and 7 vertices.

   The base is a hexagon, which has 6 sides. Faces = 1 base + 6 triangular faces = 7. Edges = 6 base edges + 6 slanting edges to the apex = 12. Vertices = 6 base corners + 1 apex = 7.
3. 2 faces (one curved, one flat), 1 edge, 1 vertex.

Rating sheet

Mark one score per criterion, then use the notes section for strengths, concerns, and recommendations. The lesson objectives are on the first page for reference.

Revision note ID	cand-b545342d	**Date**	
Subtopic	Three dimensional figures: properties of cubes, cuboids, pyramids and cones	**Rater**	

> The reference sheet includes the teacher's evaluation questions. A pack that reuses them, reworded or verbatim, may be credited for matching how the lesson was assessed — but reuse on its own does not establish taught-lesson alignment. Rate alignment on the whole of it: the objectives covered, the order taught, what the activities had the students do, the method, and the expected depth.

Criterion (select one score per row)	**1** Poor	**2** Weak	**3** Adequate	**4** Good	**5** Excellent
Objective and curriculum alignment Does the revision note cover the intended lesson objectives and curriculum target?	○	○	○	○	○
Taught-lesson alignment Does it reflect the taught lesson — its sequence, emphasis, activities, and expected depth?	○	○	○	○	○
Scope and conceptual correctness Does it stay within the expected lesson boundary and avoid conceptual/mathematical errors?	○	○	○	○	○
Student-facing clarity and usefulness Can JSS1 learners understand and use it meaningfully as take-home review material?	○	○	○	○	○

Strengths	Concerns	Recommendation

Flag any problems (optional)

Tick all that apply. These are separate from the scores above — leave blank if none apply.

☐	**Misses a required objective** — Skips something the lesson was meant to teach.
☐	**Goes beyond this lesson** — Includes material outside this lesson's scope (e.g. next week's topic).
☐	**Wrong or misleading content** — Contains a mathematical or factual mistake.
☐	**Too hard for JSS1 to follow** — Likely confusing for the students.
☐	**Other:** _____